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Research Paper

Investigating unfulfilled travel needs for people with travel-limiting conditions—Using the 2017 National Household Travel Survey (NHTS) data

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ABSTRACT

Despite the increasing awareness of unmet travel needs of people with disabilities (PWD), there is a lack of research that explores the intersectionality of disability with other aspects of an individual's identity, such as gender, race, ethnicity, employment status, income, and age. To bridge this research gap, this study investigated the disparities of unmet travel needs within specific subgroups of the disabled population using data from the 2017 National Household Travel Survey (NHTS). Studying the travel needs of PWD is crucial not only for understanding the unique challenges faced by PWD but also for uncovering potential gender and social inequities in transportation. A logit model with interaction effects was developed to identify the influential factors contributing to the decision to “reduce day-to-day travel”. By utilizing this modeling approach, the study identified key determinants that influence the occurrence of unmet travel needs among PWD and shed light on the underlying factors that contribute to the observed variations. The study uncovered the intricate relationships of gender, poverty status, race, and travel experiences among individuals with disabilities. The findings underscore the importance of exploring specific subgroups within these categories to gain deeper insights into the travel challenges and needs faced by disabled individuals. This study also reveals the potential suppression of travel desires and overlooked travel needs within the Hispanic disabled population, highlighting the necessity for targeted interventions and support to address these disparities. Policymakers are encouraged to develop strategies tailored to the specific difficulties and special travel needs of disadvantaged groups.

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1 Introduction

Travel is an integral part of people's daily lives, enabling them to connect, explore, and participate in various activities. However, individuals with disabilities often face significant challenges that affect their ability to access and enjoy these essential opportunities, including recreational activities, education, employment, and healthcare (Clery et al., 2016). In the United States, there were around 3.6 million people with travel-limiting disabilities who chose to stay home due to their

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impairments or confinement as reported in 2018 (Brumbaugh, 2018). Efforts have been made to provide modified and accessible transportation options for people with disabilities (PWD) and older individuals, such as paratransit services, accessible buses, reduced fare taxis, and “dial-n-ride” services (Wong et al., 2020; Durand and Zijlstra, 2023; Tang et al., 2018; Rahman et al., 2023), these solutions may not fully meet the diverse travel needs of the mobility disadvantaged population.

PWD face various barriers and challenges that hinder their ability to travel freely and independently. The specific types of disability and their impacts on an individual’s mobility, sensory abilities, and overall functioning can significantly influence their level of dependence on travel. Instead of traveling alone, many individuals with disabilities often rely on being passengers in personal vehicles (Brumbaugh, 2018). PWD may encounter physical barriers that include the lack of accessible transportation options, limited availability of accommodations with proper accessibility features, and challenges in accessing public spaces and attractions. Perceived barriers can be equally restrictive. These might involve societal attitudes and misconceptions about disabilities, which can lead to a lack of understanding and support from travel service providers or fellow travelers. Such attitudes can create an unwelcoming environment, discouraging individuals with disabilities from pursuing their travel aspirations.

Social equity policy underscores the importance of treating all individuals impartially, highlighting the critical necessity of ensuring fair and just opportunities for every member of society, especially for the disadvantaged population. Individuals facing transportation disadvantages often experience social exclusion and encounter more significant limitations that impede their engagement in community activities (Bascom and Christensen, 2017). Research indicates that PWD make 10–30% fewer trips than those without disabilities, especially for non-work-related purposes such as social interactions and recreational activities (Park et al., 2023). Additionally, PWD may spend more time traveling shorter distances compared with their counterparts without disabilities (Brucker and Rollins, 2016). Equity in transportation seeks fairness in mobility and accessibility to fulfill the requirements of every member of the community (Amekudzi-Kennedy, 2020). Achieving transportation equity demands addressing systemic disparities that hinder the disabled population from accessing various places and participating in essential activities (Brucker and Rollins, 2016; Fernandes-Ferreira et al., 2020; De Pascale, 2021; Lui et al., 2021; Cochran, 2020). In contrast, the lack of transportation equity prevents disadvantaged populations, especially PWD, from enjoying a high quality of life (Neven and Ectors, 2023).

Despite increased awareness of the unmet travel needs of PWD, a significant research gap remains, calling for further investigation in this area. While some studies have explored the accessibility and travel experiences of disabled individuals (Remillard et al., 2022; Cerdan Chiscano, 2021), there is a clear need for more targeted research within the disabled population itself. This entails gaining a comprehensive understanding of the diverse needs, preferences, and challenges faced by different disability groups, as well as considering various socio-demographic segments within the PWD community. Currently, there is a lack of research that explores the intersectionality of disability with other aspects of an individual’s identity, such as gender, race, ethnicity, employment status, income, and age. Investigating the interplay between disabilities and these factors can provide valuable insights into the travel experiences and specific needs of distinct subgroups within the PWD community. For example, the travel needs and experiences of disabled individuals from different ethnic backgrounds or with varying levels of income may differ significantly. Identifying and addressing these disparities can lead to the development of more inclusive and equitable transportation policies and services. By recognizing and understanding how various identities intersect with disabilities, policymakers and transportation service providers can better tailor their offerings to meet the comprehensive needs of disabled individuals.

To bridge this research gap, this study seeks to investigate the disparities of the unmet travel needs within specific subgroups of the disabled population. To achieve this goal, the study utilized data from the 2017 National Household Travel Survey (NHTS), focusing on people with travel-limiting conditions. The survey provided valuable information on personal characteristics, household attributes, land use factors, travel-related decisions, and so on. A logit model with interaction effects was developed to identify the influential factors contributing to the decision to “reduce day-to-day travel” due to their conditions, as well as to capture the disparities among different groups. The study’s findings are expected to shed light on the disparities in travel opportunities and challenges faced by PWD and its subgroups, providing valuable insights into the equity aspect of unmet travel needs. By examining the sociodemographic, environmental, and attitudinal factors that contribute to the disparities among disabled travelers, this study contributes to a deeper understanding of the unmet travel needs within specific groups of the disabled population, which can inform policymakers and practitioners in developing more targeted, inclusive strategies and policies that promote equitable access to travel opportunities for all segments of the population.

2 Literature review and hypotheses

Disability, which is considered a functional limitation or impairment that prevents an individual from fulfilling the requirements of the environment, can affect the travel choices of PWD. However, PWD are a heterogeneous group whose travel needs vary or intersect at different socio-demographic and disability characteristics (Hu, 2021; Sallis et al., 2022). In this section, we explored the literature about some key aspects of heterogeneity in disabled people’s travel needs and consequently developed some hypotheses.

2.1 Gender disparities

Generally, women tend to travel less frequently than men, and are more likely to chain their trips or make stops along the way to other destinations ([Los Angeles County Metropolitan Transportation Authority, 2019](#)). Although their trips may be shorter, they tend to spend more money on transportation ([Lecompte and Juan Pablo, 2017](#)), perhaps due to their household responsibilities that typically involve grocery shopping and taking children to school and for other out-of-home activities. However, women with developmental disabilities are more likely to have had incidents of physical and sexual assault, many of which take place or near public transit ([Iudici et al., 2017](#)). This can further discourage women with disabilities from using public transportation, leading to potentially higher mobility barriers and disparities in their travel needs. Hence, we developed the first hypothesis:

H1 Female PWD is facing more barriers in day-to-day travel than male PWD, especially with the responsibility of taking care of young children, when they live alone, or without personal vehicles.

2.2 Age disparities

The intersection of age and disability may widen disparities. Because newer generations may retain travel patterns that are different from those of their parents' generation as they grow older, exploration have been conducted to identify the potential changes in travel behavior and mobility needs of the "new older population" compared with the increasingly large ageing population. For example, [Yang et al. \(2013\)](#) examined older adults' travel time expenditure, and found that those aged 75 years or older chose transport mode primarily by trip distance/time. After a transition period into early retirement (between 65 years old and 74 years old), the trips for those aged 75–84 years and those aged 85 years or more were concentrated around noon for errands or social activity purposes.

Furthermore, multiple sociodemographic characteristics (e.g., age and gender) may intersect with a disability, creating wider disparities. Older women, for example, are generally more likely to have longer life expectancies than men, meaning that a large proportion of female seniors, particularly those over 80 years old will live alone. Older women who lost their partners tend to have lower vehicle usage and quality of life associated with car use, which can lead to more social exclusion ([Hjorthol et al., 2010](#); [Engels and Liu, 2011](#)). Since older women are more likely to give up driving as they age ([Choi et al., 2012](#)), disabled older women may have to be supported by an available companion before their travel needs can be met, and those who live alone will likely experience more difficulties in fulfilling their travel needs ([Mauch and Taylor, 1997](#); [Taylor et al., 2015](#)). Other socio-demographic factors such as income, education, and race may intersect with a disability status, limit the transportation options, and potentially widen disparities. Hence, the second hypothesis is:

H2 Older PWD is more likely to reduce day-to-day travel, as they might feel social isolation as they get older, especially when they live alone or in poverty.

2.3 Race and ethnic

In some way, people of color experience inadequate transportation accessibility due to the variations in travel behavior arising from disparities in employment and educational opportunities, as well as familial and informal community commitments ([Karner and Niemeier, 2013](#)). Literature found that Hispanic women with disabilities have significantly shorter commutes than comparable white men ([Wong et al., 2020](#)). Hispanic/Latino adults appear to have relatively low leisure-time physical activity and low rates of meeting physical activity guidelines ([Sallis et al., 2022](#)). The third hypothesis is:

H3 Colored or Hispanic PWD perceives more barriers in traveling since they are socially disadvantaged groups. Those colored or Hispanic PWD with poverty status or low education are more vulnerable.

2.4 Disability characteristics

Aside from intersection of various socio-demographic attributes and disabilities, the type and seriousness of disabilities can affect the PWD's travel needs and choices. The differences in travel needs exist among people with different types of disabilities, as PWD may require different accessible transportation infrastructure, such as wheelchair ramps or elevators, or communication support or information in accessible formats ([Mulcahy et al., 2022](#)). Some disabled individuals prefer to travel independently, while others rely on companions or accessible transportation services. For individuals with visual impairments, navigating unfamiliar environments and accessing transportation services without readily available schedule information can be difficult ([Wong, 2018](#)). Additionally, the duration or length of a disability (how long the disability has persisted) can significantly influence the travel choices and needs of PWD. Individuals with long-term disabilities tend to be more aware of options addressing their travel needs, and are likely to develop effective transport strategies ([Henly and Brucker, 2019](#)). In contrast, those newly or temporarily disabled may lack familiarity or confidence in using assistive devices, resulting in a reluctance to travel for most purposes ([Ward et al., 2017](#)). Accordingly, we developed the fourth hypothesis:

H4 PWD with different lengths of disabled status may feel differently about travel needs fulfillment. Lifetime disabled people may not consider that their day-to-day travel has been reduced, as they might have adapted their lifestyles to accommodate their disability.

2.5 Summary

The studies presented suggest significant heterogeneity in the travel needs of PWD, which are attributable to various socio-demographic and disability characteristics. These factors can lead to distinct travel needs among PWD, necessitating the examination of the unfulfilled travel needs among PWD.

3 Data description

To explore the equity aspect of the unmet travel needs of PWD, we utilized data from the 2017 NHTS. The 2017 NHTS provides comprehensive information about American personal, household, and travel characteristics (McGuckin et al., 2018). The survey contained information about 264 234 individuals from 129 696 households across the United States. To focus specifically on PWD, we extracted data from respondents who reported having travel-limiting disabilities in the survey. For the purpose of this study, respondents with travel-limiting disabilities were defined as individuals with a temporary or permanent condition or handicap that makes it difficult for them to travel outside of their homes. It is important to note that some respondents might have disabilities that do not limit their travel, and thus, they were not included in this analysis. Additionally, the sampling method employed by NHTS relied on an address-based frame, meaning that individuals living in nursing homes or other group quarters were not part of the study's sample. Furthermore, our analysis focused solely on adults aged 18 and above with a disability. After excluding missing values, the final sample size used in this study comprised 23 644 observations.

To ensure meaningful analysis and comparison across different life stages, we categorized respondents into the following age groups: 18–24, 25–44, 45–64, 65–74 (young-old), and 75 and older (old-old). These groupings were selected to capture significant phases in the life course, such as young adulthood, mid-career, pre-retirement, retirement, and advanced age, which correspond to distinct transportation needs and challenges. The inclusion of the young-old and old-old categories allows for a more nuanced understanding of how aging might affect travel behavior and travel needs (Glasgow and Blakely, 2000; Sadehghvaziri and Tawfik, 2020). These categorizations are consistent with prior research on travel behavior and age-related mobility patterns (Buehler et al., 2020; Uddin et al., 2023; Pucher et al., 2011).

To gain insights into how disability status might influence daily trips, all disabled persons were asked to indicate if their medical condition resulted in “reduced day-to-day travel”, using a binary category of “yes” or “no”. In this study, this variable serves as an indicator of unfulfilled travel needs, with individuals who chose “yes” being those who experience unmet travel needs due to their disability, compared with others. Overall, 72.4% of disabled respondents reported that they had reduced day-to-day travel due to their disability.

For the purposes of this study, “unmet travel needs” are conceptualized as situations where individuals with disabilities are hindered from making trips they desire or require for full societal participation, often because their travel-limiting conditions interact with systemic environmental, economic, or social barriers, leading to suppressed travel. To empirically measure this multifaceted concept using the 2017 NHTS data, we operationalize “unmet travel needs” based on the respondents’ self-reported reduction in travel. Specifically, the binary response (“yes” or “no”) to the question of whether their medical condition resulted in “reduced day-to-day travel” is used as the direct empirical indicator. An affirmative response (“yes”) is interpreted as evidence of an individual experiencing an unfulfilled travel need, forming the basis for our dependent variable.

The descriptive statistics of the data used in this study are presented in Table 1.

4 PWD and the general population

To better understand the individual characteristics of PWD, the crosstabulations of PWD and the adult population of the United States were plotted by age, gender, and income. Fig. 1 shows the comparison by age distribution. It is observed that a lower percentage of the younger age groups (i.e., 18–24 years, 25–44 years, and 45–64 years) have travel-limiting disabilities, while a higher percentage of older adults (i.e., 65–74 years and 75 + years) are disabled. This suggests that as individuals age, they may be more likely to have disabilities or limitations that affect their daily activities and functioning.

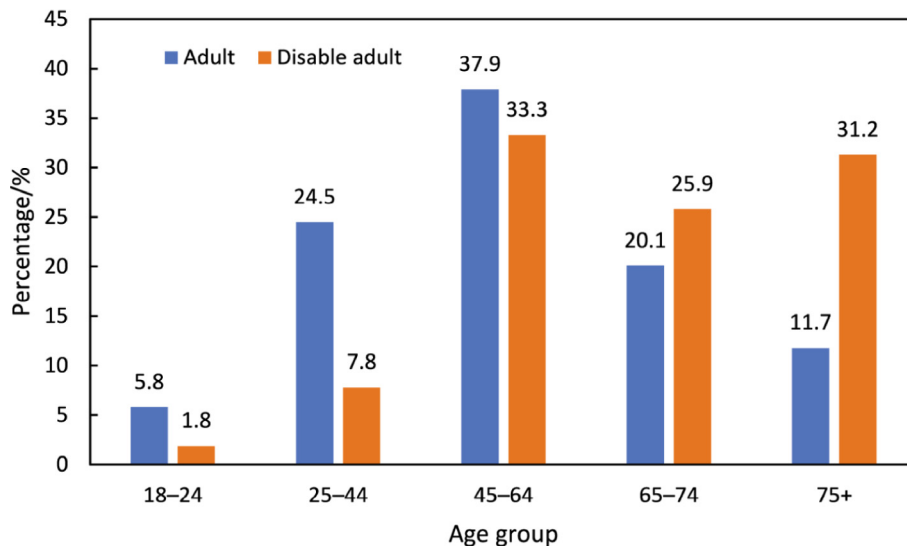
Fig. 2 displays the gender distribution of PWD compared with the general population. The data reveal a notable gender pattern within the disabled adult demographic. A slightly higher proportion of disabled adults, specifically 59.1%, are female. This surpasses the overall percentage of females in the adult population, which stands at 53.3%. In contrast, the proportion of disabled male adults is slightly lower, accounting for 40.9%. This figure is marginally beneath the percentage of male adults in the general population, which is recorded at 46.7%. The disparities observed in the gender distribution of disabled adults compared with the general population underscore the importance of conducting nuanced analyses that account for both disability and gender.

Fig. 3 highlights the relationship between income level and travel-limiting disabilities among PWD compared with the general population. It reveals that a significant percentage of PWD classified as poor have travel-limiting disabilities. Specifically, while one-quarter of American adults live in households with incomes under 35 000 USD, over half of disabled adults reside in such households. This suggests that income level may be associated with the likelihood of having travel-limiting disabilities, therefore leading to unfulfilled travel needs.

Table 1

Sample characteristics.

Variable	Description	Frequency	Percentage/ %
Age group	18–24	436	1.8
	25–44	1839	7.8
	45–64	7865	33.3
	65–74	6117	25.9
	75+	7387	31.2
Gender	Male	9673	40.9
	Female	13 971	59.1
Race	White	19 582	82.8
	Non-White	4062	17.2
Ethnic group	Non-Hispanic	22 350	94.5
	Hispanic	1294	5.5
Education level	Less than a high school graduate	9853	41.7
Poverty status (household income <35 000 USD)	Poverty	11 468	48.5
Household members	Only adult in households	6949	29.4
Has young children (0–4 years)	Yes	574	2.4
Home type	Owned	17 189	72.7
	Rented	6455	27.3
Employment status	Worker	2944	12.5
	Non-worker	20 700	87.5
Household vehicle ownership	Zero vehicle	2679	11.3
Built environment	Urban	17 842	75.5
	Rural	5802	24.5
Residential density in home location (housing units per square mile in the census block group)	0–99	5380	22.8
	100–499	4551	19.2
	500–999	3191	13.5
	1000–1999	4586	19.4
	2000–3999	4057	17.2
	4000–9999	1430	6.0
	10 000–24 999	329	1.4
	25 000–999 999	120	0.5
Disability type	Vision disabled	364	1.5
	Wheelchair user	3 172	13.4
Disability length	<6 months	1 897	8.0
	>6 months	19 850	84.0
	Lifetime	1897	8.0

**Fig. 1** Age distributions of adults and disabled adults in the United States.

As shown in Fig. 4, white individuals make up a slightly smaller proportion of the disabled adult population (82.8%) compared with their representation in the general adult population (84.6%). In contrast, black individuals are significantly over-

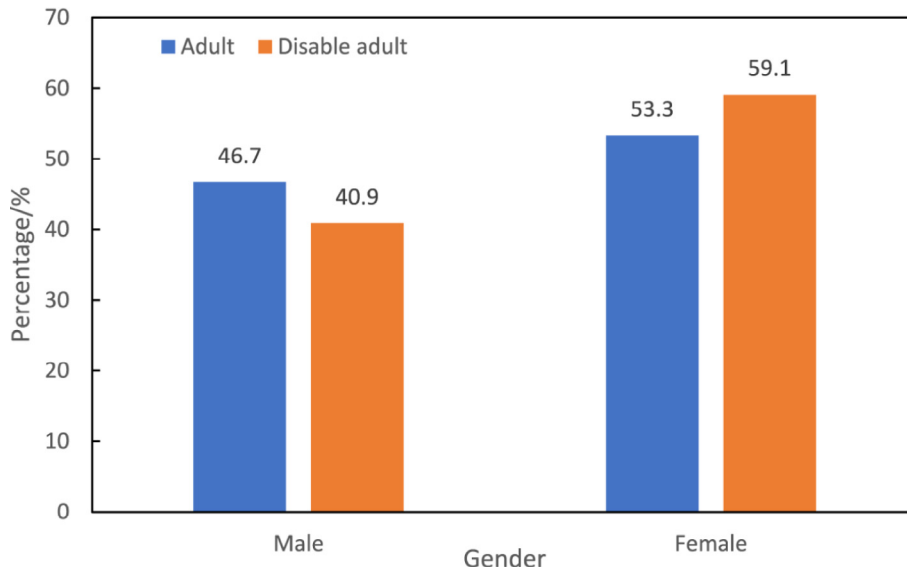


Fig. 2 Gender distribution of adults versus disabled adults in the United States.

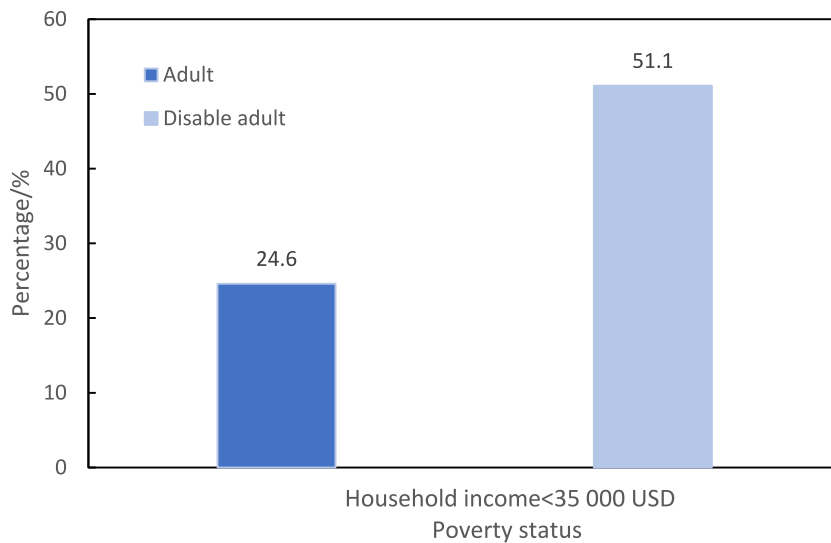


Fig. 3 Poverty status of adults and disabled adults in the United States.

represented among PWD, constituting 10.3% of the disabled population versus 7.3% of the general population. This overrepresentation suggests that black adults are more likely to experience travel-limiting disabilities, which could be due to a variety of systemic factors, including economic disparities, healthcare access, and environmental conditions. The Asian population is underrepresented among PWD (2.3%) compared with their proportion in the general population (4.6%), which cultural differences might influence in reporting disabilities or variations in health outcomes. Additionally, American Indian or Alaska Native individuals are more represented among PWD (1.0%) than in the general population (0.6%), pointing to specific challenges within this community.

Fig. 5 focuses on the Hispanic/Latino population, revealing that they are slightly more represented among PWD (8.1%) compared with their proportion in the non-Hispanic/Latino population (7.4%). This small disparity suggests that while Hispanic/Latino individuals are proportionately represented, they still face barriers that could contribute to disabilities, such as limited access to healthcare, language barriers, and economic challenges.

Fig. 6 presents a stark contrast between the employment status of PWD and the general adult population of the United States. The data indicates significant disparities that underline the challenges faced by PWD in the labor market. Among the general adult population, 55.0% are employed. However, this figure drops sharply to just 12.5% among PWD. Conversely,

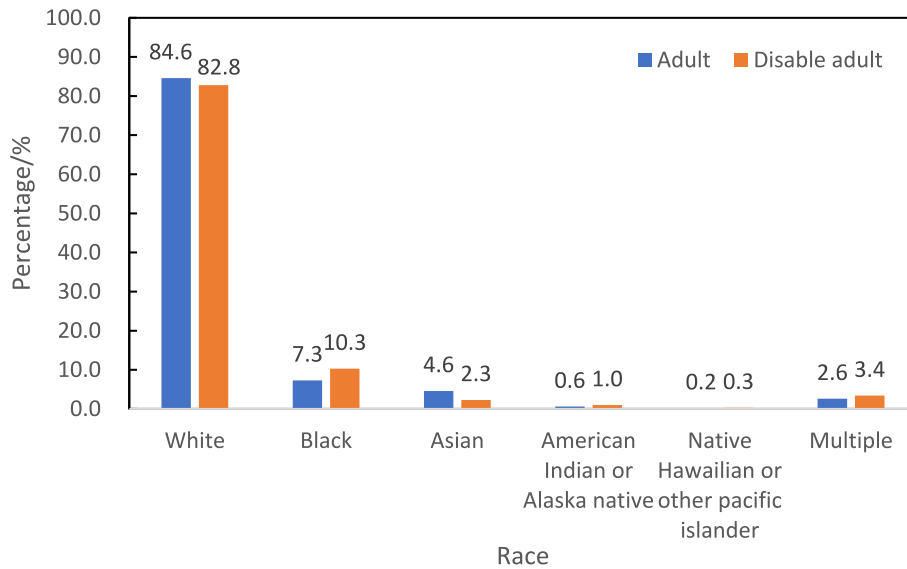


Fig. 4 Race of adults and disabled adults in the United States.

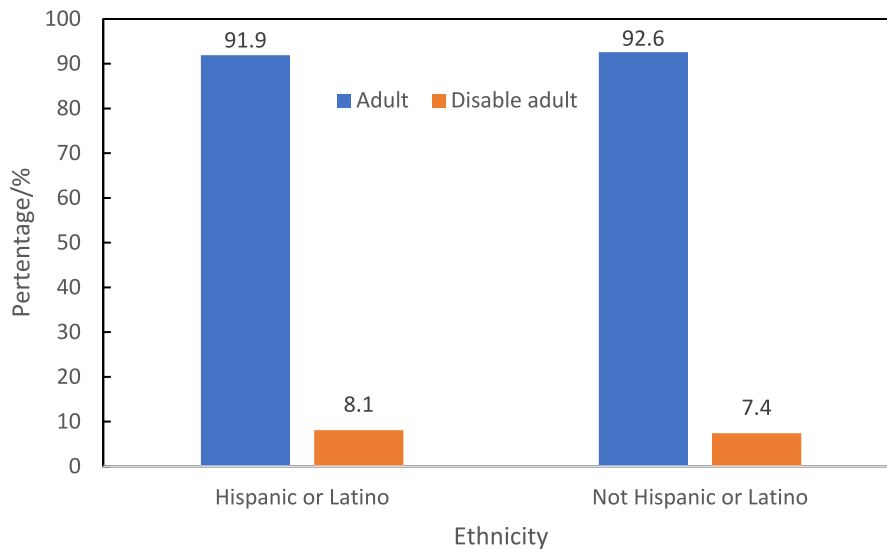


Fig. 5 Ethnicity of adults and disabled adults in the United States.

87.5% of PWD are not employed, compared with 45.0% of the general adult population. This disparity suggests that PWD are more than twice as likely to be unemployed or out of the labor force altogether. The low rate of employment and high rate of non-employment among PWD may contribute to economic disadvantages and exacerbate other challenges, such as access to healthcare and transportation.

Fig. 7 illustrates the differences in the educational attainment between PWD and the general adult population of the United States. A notable 10.8% of PWD have less than a high school education, compared with only 4.1% of the general adult population. This disparity indicates that PWD are more than twice as likely to have lower educational attainment, which can limit their access to employment opportunities and other resources. The percentage of PWD who have completed high school (31.5%) is significantly higher than the percentage of the general population (20.5%). This suggests that while a substantial portion of PWD attains a high school diploma, fewer can pursue further education, possibly due to financial, accessibility, or health-related barriers. A similar proportion of PWD (32.5%) and the general population (30.0%) have attended some college or obtained an associate's degree. However, the data also suggest that many PWD may face challenges completing these programs or advancing their education. A striking difference is observed at the bachelor's degree level, where only 13.4% of PWD have obtained a degree, compared with 24.1% of the general population. This gap highlights the signif-

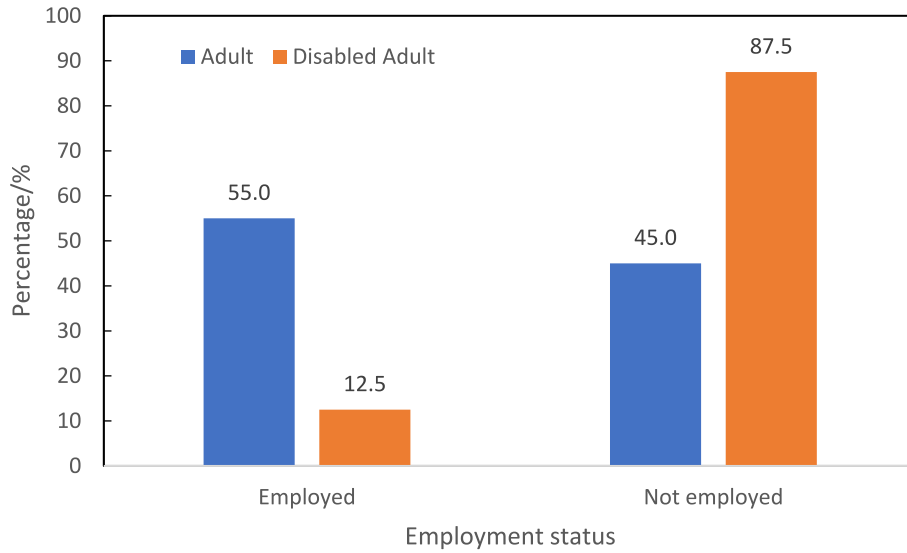


Fig. 6 Employment status of adults and disabled adults in the United States.

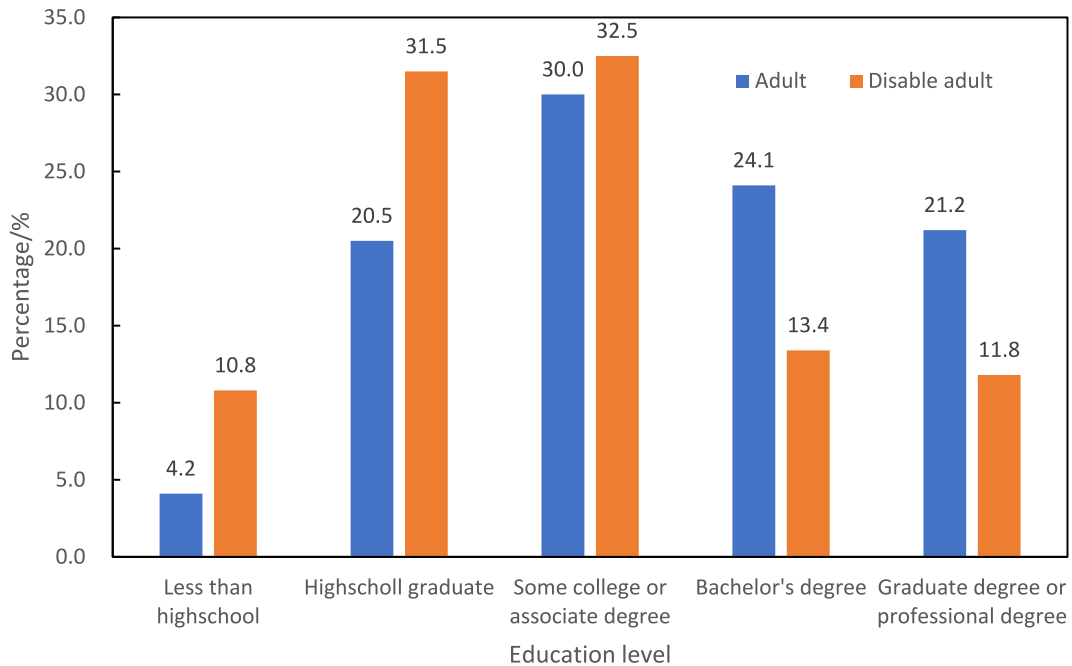


Fig. 7 Education levels of adults and disabled adults in the United States.

icant barriers PWD face in accessing and completing higher education, which could include financial constraints, lack of accessible educational resources, or health-related challenges that interfere with their studies. The disparity continues at the graduate level, with only 11.8% of PWD attaining a graduate or professional degree, compared with 21.2% of the general population. This further emphasizes PWD's challenges in pursuing advanced education, which can impact their ability to secure higher-paying jobs and achieve economic stability. The data reveal significant disparities that highlight the challenges PWD face in achieving higher education levels, which can, in turn, impact their economic opportunities and quality of life.

5 Methodology

To examine the influential factors contributing to unfulfilled travel needs among PWD, a binomial logistic regression model with interaction effects was utilized in this study. To gain a comprehensive understanding of the complexities

involved, the interaction effects were introduced in the model between the main variables of interest, such as gender, age, ethnicity, and disability type, and various other socio-demographic characteristics and land use factors. This approach allows for the assessment of potential sources that contribute to the disparities observed in the unfulfilled travel needs among different subgroups of PWD.

The binomial logistic regression model with interaction effects can be expressed as:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum(\beta_j \mathbf{X}_{ij}) + \sum(\gamma_{jk} \mathbf{X}_{ij} \mathbf{X}_{ik}), \quad (1)$$

where p_i is the probability of respondent i experiencing reduced travel, β_0 is the intercept term, $\mathbf{X}_{ij}$ is the vector of the j th predictor variable associated with the reduced travel, β_j is the coefficient of the corresponding parameter, $\mathbf{X}_{ik}$ is the vector of the k th independent variable, and γ_{jk} is the coefficient associated with the interaction between the j th and k th predictor variables.

In the model, the dependent variable is a binary outcome of whether or not PWD reduced their day-to-day travel due to their disabilities, which serves as a proxy measuring unfulfilled travel needs. The independent variables used include the main variables of interest as discussed in the hypotheses, along with various other factors such as the socio-economic and demographic variables, built environment attributes, and disability characteristics. The socio-economic and demographic characteristics include respondents' age, gender, race, ethnicity, employment, education level (if less than high school), poverty status (income <35 000 USD), household size (if living alone in the household), vehicle ownership, housing type, and having a young child in the household. Also, the built environment characteristics used are the level of residential urbanity (urban or rural) and density, while disability length (<6 months, >6 months or lifetime) and disability type (wheelchair user or vision disabled) were the disability characteristics included in the model.

The independent variables were coded as dummy variables to allow for the representation of their categorical nature (e.g., age group and residential density) and for practical usefulness. All the independent variables were included in the model all at once, and the backward stepwise selection method was used to remove variables that do not contribute significantly to the prediction of the outcome, thereby creating a simpler and interpretable model. The threshold for removal was at the 95% confidence level (i.e., $-1.96 > z\text{-value} < 1.96$).

6 Results

Table 2 presents the results of the heterogeneous model (the model with main and interaction effects). This table displays the coefficients (B) and significance levels (P) for significant variables. The result table highlights only the significant variables and their corresponding coefficients, which are the primary focus of our analysis. However, the final model also includes some variables that were found to be statistically insignificant. These variables remain in the final model because it was constructed using the backward stepwise selection method, where variables are systematically removed based on

Table 2
Logistic regression results of PWD's unfulfilled travel needs.

Indicator		B	P
Socio-demographic	Age 25–44 (reference: all other age groups)	0.234	**
	Hispanic (reference: non-Hispanic)	−0.801	***
Socio-economic	Unemployed (reference: employed)	0.737	***
	Only adult	0.195	***
Disability status	Disabled status <6 months	1.565	***
	Disabled status >6 months	1.021	***
	Wheelchair user (reference: not wheelchair user)	0.263	***
Gender interaction effect	(Disabled status >6 months) × female	0.266	**
	(Disabled status lifetime) × female	0.260	*
	Unemployed × female	−0.196	**
	(Education less than high school) × female	−0.174	***
Age interaction effect	(Disabled status lifetime) × (age 65–74)	0.474	***
	(Disabled status lifetime) × (age 75+)	0.478	***
	Poverty × (age 45–64)	0.221	**
	Poverty × (age 65–74)	0.270	**
	Poverty × (age 75+)	0.236	**
	(Only adult) × (age 65–74)	−0.315	***
	(Only adult) × (age 75+)	−0.296	***
	(Wheelchair user) × (age 65–74)	0.336	***
Hispanic interaction effect	(No car) × Hispanic	−0.438	**
	(Home rented) × Hispanic	−0.801	***
	(Education less than high school) × Hispanic	−0.801	***
Constant		−1.520	***

B : The estimated log-odds (logit) coefficient.

Level of significance: * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$. McFadden R -square: 0.026 99.

their statistical significance. An insignificant variable in the model indicates that, after accounting for the other variables in the model, changes in this predictor do not significantly influence the outcome variable. Specifically, this means that when the predictor variable is set to zero, the effect on the outcome does not differ significantly from the model's constant, which represents the baseline level of the outcome variable when all predictors are zero.

The results from the main effects indicate that PWD who use wheelchairs tend to have unfulfilled travel needs. The use of a wheelchair indicates a disability more related to mobility limitation than other forms of disability, and those in this category may have to rely fully on others to meet their travel needs. Also, PWD who have not had their disability throughout their lifestyle were more likely to have unfulfilled travel needs, perhaps because of their limited experience in navigating through their disability (Brucker and Rollins, 2016). Unlike newly disabled individuals who may have had their life and lifestyles significantly altered by a disability, those who have had a disability from birth may have structured their life around the disability and do not have to alter any “former lifestyle”, contributing to lower unfulfilled needs for “lifetime” PWD.

Regarding socio-economic and socio-demographic characteristics, PWD who were aged 25–44, non-Hispanics, living alone, and unemployed were more likely to have unfulfilled travel needs than the older population. This could be attributed to the tendency of younger individuals to engage in more social and recreational activities, unlike the older population. Hispanic PWD were less likely to have unfulfilled travel needs despite being considered a transportation-disadvantaged group, as they tend to have strong communities where their travel needs can be met by other community members (Francis et al., 2018). Expectedly, those who live alone tend to have unfulfilled travel needs since there are no easily accessible family members in the household who could offer help when the need arises. Interestingly, those who are unemployed tend to have unfulfilled travel needs. However, this might be an indication of the seriousness of their disability, which prevents them from being employed or easily employable. It should be noted that the main effects of variables such as poverty status, gender, car ownership, and built-environment variables were not statistically significant in the model.

The results of the interaction effects in Table 3 showed that some variables that were not significant as a main effect became significant when interacting with other variables. For instance, female PWD and those who have had a disability since their birth did not predict unfulfilled travel needs, but the effect of the interaction of both variables positively predicted unfulfilled travel needs. Also, the interaction between having a disability for more than six months and being female showed a positive effect on unfulfilled travel needs. Among the unemployed PWD subgroup, females are less likely to predict unfulfilled travel needs than their male counterparts. Although these effects may indicate that there are more travel barriers for disabled females, the low-educated female PWD were less likely to have unfilled travel needs. While the reasons for these effects are unclear, we speculate that these female groups tend to perceive no unfulfilled travel needs, as stay-at-home mothers or wives.

Including these insignificant variables in the final model allows us to capture potential interactions with significant variables, avoid omitted variable bias, and ensure that the model remains theoretically sound and comprehensive. In Eq. (1) presented in Section 5, with predictors X_{ij} and X_{ik} and the interaction term which is the product of X_{ij} and X_{ik} , the interaction effect represents that the effect of X_{ij} depends on the value of X_{ik} and vice versa, which equals $\Sigma(\beta_j + \gamma_{jik})$. The insignificant variables in the main model β equals to 0. However, while there is no significant overall effect of either gender or education level individually, our model indicates a crossover interaction between them. Specifically, the effect of gender on the perception of unfulfilled travel needs is negative and varies depending on the level of education, and vice versa. This interaction suggests that the influence of female PWD's perception on unfulfilled travel needs is contingent upon the PWD's low education level, and the impacts of low education on these perceptions are similarly dependent on gender.

The interaction between Hispanics and other socio-economic and demographic characteristics like home, vehicle ownership, and education were not associated with unfulfilled travel needs. Specifically, Hispanic PWD, particularly those with no car accessibility, living in rented homes, and lower educational levels were all less likely to consider themselves as having

Table 3
Interpretation of the interaction effects.

Effect	Indicator 1	Indicator 2	Log-odds ratio calculation	Log-odds ratio
Gender interaction effect	Disabled status >6 months	Female	1.021 + 0.266	1.287
	Disabled status lifetime	Female	0.260 + 0	0.260
	Unemployed	Female	0.737 – 0.196	0.541
	Education less than high school	Female	–0.174 + 0	–0.174
Age interaction effect	Disabled status lifetime	Age 65–74	0.474 + 0	0.474
	Disabled status lifetime	Age 75+	0.478 + 0	0.478
	Poverty	Age 45–64	0.221 + 0	0.221
	Poverty	Age 65–74	0.270 + 0	0.270
	Poverty	Age 75+	0.236 + 0	0.236
	Only adult	Age 65–74	0.195 – 0.315	–0.120
	Only adult	Age 75+	0.195 – 0.296	–0.101
	Wheelchair user	Age 65–74	0.263 + 0.336	0.599
	Wheelchair user	Age 75+	0.263 + 0.336	0.599
Hispanic interaction effect	No car	Hispanic	–0.801 – 0.438	–1.239
	Home rented	Hispanic	0.431 – 0.801	–0.370
	Education less than high school	Hispanic	0.379 – 0.801	–0.422

unfulfilled travel needs. Literature suggests that due to compounded challenges such as language barrier, immigration status, and specific needs, PWD in the Hispanic community tend to trust their neighbors and other community members more than service providers when seeking assistance, such as finding assistive technology or housing (Francis et al., 2018). Under these circumstances, they may not perceive the unfulfilled travel needs while the travel barriers still exist. This complex situation, where reported needs may not reflect true underlying deprivation due to cultural factors, trust issues, or adaptation to systemic barriers, is critical for the broader debate on transport equity. It highlights the inadequacy of demand-responsive planning alone and underscores the need for proactive, culturally competent outreach and service design to ensure genuine social inclusion for such communities, moving beyond simply addressing expressed travel demand to uncovering and tackling suppressed mobility.

Age-related interaction effects revealed varying effects. Older PWD (65 years and above), who are poor, use a wheelchair or have been disabled from birth, tend to have unfulfilled travel needs. Surprisingly, older PWD who live alone are less likely to have unfulfilled travel needs. Since age was not significant in the main effect model, the significant and varying effects of the interactions of age groups and other groups are indicative of much higher heterogeneity among PWD.

To enhance the reliability and stability of the main findings, a series of robustness checks was performed on the logistic regression model presented in Table 2. These checks involved re-estimating the model under various alternative specifications for key independent variables and assessing the consistency of the primary relationships. Specifically, variations included employing alternative age group categorizations, using different poverty thresholds, re-categorizing education levels, and re-introducing variables that were initially excluded during the backward stepwise selection process. Overall, the core findings regarding the factors influencing unfulfilled travel needs among PWD demonstrated strong robustness across these analyses.

Table 4 presents the results of robustness checks performed on the primary logistic regression model, aiming to verify the stability and reliability of the key findings. This table displays the coefficients (*B*) and significance levels (*P*) for central variables from the main model (see Table 2) alongside those from several alternative model specifications. These alternative specifications included variations in age group categorization, poverty threshold definition, education grouping, and the re-inclusion of a previously excluded non-significant variable. The consistent signs and significance levels of the key predictors across all tested models, despite minor fluctuations in magnitude, confirm the robustness of the identified relationships, thereby reinforcing the reliability and generalizability of the conclusions regarding unfulfilled travel needs among PWD.

As shown in Table 4, the coefficients for primary predictors, including socio-demographic factors, like Hispanic ethnicity and employment status, and disability characteristics, such as wheelchair use and disability duration, consistently retained their statistical significance and direction of effect across all tested models. For instance, the negative association between Hispanic ethnicity and the likelihood of reporting reduced travel remained robust, as did the positive association for unemployed PWD. Similarly, the key interaction effects, such as the mitigating influence of low education on reduced travel for

Table 4

Robustness check—key coefficients from alternative logistic regression models for unfulfilled travel needs.

Variable		Main model		RC1: adult age group		RC2: adult poverty thresh		RC3: adult education group	
		<i>B</i>	<i>P</i>	<i>B</i>	<i>P</i>	<i>B</i>	<i>P</i>	<i>B</i>	<i>P</i>
Main effects	Age 25–44	0.234	**	—(dummy group)	—	0.23	**	0.235	**
	Hispanic	−0.801	***	−0.798	***	−0.805	***	−0.800	***
	Unemployed	0.737	***	0.740	***	0.735	***	0.738	***
	Only adult	0.195	***	0.192	***	0.197	***	0.194	***
	Disabled status <6 months	1.565	***	1.560	***	1.568	***	1.564	***
	Disabled status >6 months	1.021	***	1.025	***	1.020	***	1.022	***
	Wheelchair user	0.263	***	0.268	***	0.264	***	0.262	***
Key interaction effects	(Disabled status >6 months) × female	0.266	**	0.265	**	0.267	**	0.265	**
	(Education less than high school) × female	−0.174	***	−0.172	***	−0.175	***	Varies (variable changed)	Varies
	(Disabled status lifetime) × (age 65–74)	0.474	***	Adjusted for new age	Adjusted	0.470	***	0.473	***
	Poverty × (age 65–74)	0.270	**	Adjusted for new age	Adjusted	0.272	**	0.270	**
	(Only adult) × (age 75+)	−0.296	***	Adjusted for new age	Adjusted	−0.295	***	−0.296	***
	(Wheelchair user) × (age 65–74)	0.336	***	Adjusted for new age	Adjusted	0.335	***	0.336	***
	(No car) × Hispanic	−0.438	**	−0.435	**	−0.440	**	−0.437	**
Constant		−1.52	***	−1.515	***	−1.522	***	−1.518	***

B: The estimated log-odds (logit) coefficient.

Level of significance: **P* < 0.1; ** *P* < 0.05; *** *P* < 0.01.

female PWD, and the complex age-related dynamics observed for poverty and living alone, consistently maintained their significance and direction. Minor variations in the precise magnitudes of coefficients were observed across these alternative specifications, but these did not alter the qualitative conclusions or the substantive interpretation of the effects. This consistency across varied model specifications strengthens the confidence in the reliability and generalizability of the identified relationships between socio-economic factors, disability characteristics, and perceived unfulfilled travel needs among PWD.

7 Hypotheses review and discussion

The findings regarding the association between being a female and unfulfilled travel needs confirm our first hypothesis. Although being a disabled female was insignificant as the main effect, the gender factor gains significance when interacting with other variables. It reveals complex travel needs among female PWD, particularly emphasizing the significance of interaction effects and the role of gender in influencing perceived unmet travel needs. The intersection between disabled females and longer duration among females indicates the tendency to a higher likelihood of unmet travel needs. Specifically, females with disabilities lasting more than six months are more prone to experiencing unfulfilled travel needs. This might suggest that, over time, disabled females may have experienced some form of social exclusion or discrimination because of their disability. For example, disabled females may face additional challenges in accessing transportation due to caregiving responsibilities or expectations regarding their mobility (Currie, 2011). It is also possible for disabled females to have become more aware of their limitations or discrimination against them over time, and are consequently more discouraged about traveling. Among the subgroup of unemployed PWD, females are less likely to predict unfulfilled travel needs compared with their male counterparts. This suggests a potential gender-based variation within the unemployed PWD category, with females facing fewer barriers to travel despite their unemployment status. Despite potential barriers indicated by other interactions, a noteworthy finding is that low-educated female PWD were less likely to have unfulfilled travel needs. This protective effect may suggest unique travel experiences for stay-at-home mothers or wives (Wang and Xu, 2020).

Our second hypothesis concerns the older population, and was also confirmed by our results. Like the effect of females on unfulfilled travel needs, the effect of disabled older adults was not significant as a main effect. Still, it was significant at the intersection with other socio-demographic factors. Poor, lifelong disabled older adults, for example, were more likely to have unfulfilled travel needs. Also, other groups of disabled older adults were less likely to have unfulfilled travel needs, suggesting heterogeneity within the group. These findings highlight the complexity and subtleties within different age groups and social statuses among PWD.

Concerning the third hypothesis, the negative effects of the Hispanic group contradict the hypothesis. Moreover, interaction with other socio-demographic characteristics shows similar effects and suggests that Hispanic individuals with disabilities do not have unfulfilled travel needs compared with the non-Hispanic population. The interaction effects reveal that specific disadvantaged subgroups within the Hispanic population with disabilities exhibit distinct travel patterns and needs.

Like the first two hypotheses, the fourth hypothesis was confirmed by our results in the main effect model, but by the results from the interaction effect model. Newly disabled individuals were likely to have unfulfilled travel needs, but only at the intersection with females and older adults. This underscores the need for increased attention to the challenges of females and older adults with travel-limiting disabilities.

It is important to note that an indication of having reduced travel does not necessarily indicate that individuals had fewer trips or encountered fewer barriers. On the contrary, it may reflect the suppression of their travel needs due to systemic inequities faced by disabled individuals when it comes to traveling. For example, the same constraint can lead to non-participation or limit the performance of activities and affect frequency or satisfaction (Small et al., 2012).

It should be noted that there are two situations in which respondents would consider their day-to-day travel as not reduced. The first situation is when individuals have high travel demand, but all of their travel needs are met. In this case, their travel needs are effectively fulfilled, resulting in a perception of reduced day-to-day travel. The second “not reducing” situation happens when individuals have a low desire to travel outside. However, this does not indicate that their travel needs are met. On the contrary, it may indicate social inequities faced by PWD. In this context, individuals may have reduced day-to-day travel not due to fulfilled travel needs but due to barriers, constraints, or social factors that limit their desire or ability to engage in travel activities. This highlights the need to consider systemic inequities and accessibility issues impacting travel choices and opportunities for PWD.

The research also implies that “access” in transportation must be understood more broadly than mere physical accessibility of vehicles or infrastructure. While essential, physical access alone does not guarantee equitable mobility if economic barriers remain prohibitive, if services are not culturally appropriate or trustworthy (as potentially with the Hispanic PWD population), or if the psychological toll of navigating unwelcoming environments leads to travel avoidance. A holistic and socially attuned understanding of access is imperative for fostering genuine social inclusion.

8 Policy implications

While efforts have been made to provide modified and accessible transportation options for PWD and older individuals, such as paratransit services, accessible buses, reduced-fare taxis, and door-to-door transportation services (Wong et al.,

2020; Durand and Zijlstra, 2023; Tang et al., 2018; Rahman et al., 2023), these solutions may not fully meet the diverse travel needs of the complex PWD population.

The results from the interaction effects indicate that specific subgroups of female PWD may face unique challenges and limitations in their travel experiences, such as females with longer duration of disability or those who are unemployed. Longer length of disability will incur higher transportation costs on time and finance due to the limited mobility options and specialized transportation needs. Disabled females may experience social exclusion or discrimination based on both their disability and gender, leading to compounded experiences of marginalization. Policies should focus on understanding and addressing the specific needs and barriers faced by these subgroups, creating a safer and more inclusive transportation environment for female PWD and all users. This can involve providing targeted transportation services, accessibility improvements, and social support to empower and enable independent travel. Policymakers should consider taking an intersectional approach to address sexual harassment by considering the unique experiences and challenges faced by disabled females who may experience multiple forms of discrimination (Noor and Iamtrakul, 2023). For instance, some cities have experimented with demand-responsive transit services specifically catering to women's safety concerns during off-peak hours, or initiatives focused on gender-sensitive design in transport hubs, drawing lessons from programs in India and Barcelona, Spain (Giuffrida et al., 2025). Similarly, anti-harassment campaigns on public transport, such as those implemented in Japan, India, and Mexico City in Mexico, sometimes include specific components addressing the vulnerability of women with disabilities (Borker, 2024; Aguilar, 2021). Implementing such gender-focused transportation policies can present challenges, including securing sustained funding for specialized services, ensuring that targeted interventions do not inadvertently lead to segregation or stigma, effectively training transport staff in gender sensitivity and disability awareness, and robustly monitoring and evaluating program effectiveness to adapt to evolving needs and contexts.

In the meantime, addressing finance-related travel barriers for the PWD community is urgent. Policies should aim to alleviate the financial burden and economic disparities faced by this subgroup. Providing transportation subsidies or vouchers for individuals with disabilities who are unemployed or have longer durations of disability can be an effective way to address their transportation needs (Blumenberg and Agrawal, 2014). This policy directly targets the mechanism of economic constraint, a key reason identified for unfulfilled travel, which disproportionately affects PWD who are out of the workforce by limiting their ability to afford transportation. By alleviating this financial burden, such subsidies can enhance access to essential services and opportunities, thereby addressing a significant contributor to their potential social exclusion. These subsidies can help alleviate the financial burden of transportation costs and increase access to essential services, such as medical appointments, job interviews, and social activities. By making transportation more affordable and accessible, these subsidies can help improve the quality of life for individuals with disabilities and enhance their ability to participate fully in their communities. However, subsidies may not cover the full cost of transportation, leaving individuals with disabilities to bear some of the financial burden (Currie and Delbosc, 2010). Wong et al. (2020) suggested that, for the full implementation of the taxi fare subsidy scheme, a phased approach could be considered for the elderly, with different age groups receiving varying levels of subsidy. This strategy may be more effective in encouraging older individuals to use taxis for their travel needs. Expanding on the policy suggestion, a step-wise subsidy scheme could also be applied to individuals with disabilities based on the duration of their disability and their poverty levels. For example, those with temporary disabilities or who are newly disabled could receive higher subsidies initially, gradually decreasing as the duration of their disability increases. Additionally, different types of vouchers for specialized transportation services could be provided to people with different types of disabilities, ensuring that their specific needs are met effectively. In addition, there may be a stigma associated with using transportation subsidies, leading some individuals with disabilities to avoid using them. Especially when the application process for subsidies may be complex or difficult to navigate, particularly for individuals with cognitive or communication disabilities (Neven and Ectors, 2023). Policymakers and transportation providers should consider simplifying the application processes and ensuring the privacy of the applicants.

Ensuring that public transportation services are accessible to individuals with disabilities is essential for promoting equality, human rights, safety, social inclusion, and economic opportunities for all members of society, including those with disabilities (de Pascale et al., 2021). This includes providing accessible vehicles, ramps, and platforms, as well as offering assistance services for those who need them. Accessible transportation enables individuals with disabilities to travel independently, participate in social activities, access essential services, and pursue employment and educational opportunities. Accessible public transit in general can be a more affordable transportation mode compared with door-to-door transportation services. However, the affordability of public transit for PWD may vary depending on factors such as the availability and accessibility of public transit services in their area, the extent of their disability, and any additional costs associated with accessing public transit (Silcock, 1986).

It is worth noting that the findings indicate that Hispanic individuals with disabilities, particularly those with no car accessibility, living in rented homes, and lower educational levels, may have unrealized mobility or suppressed mobility. Their desire to have trips and activities is prevented by various barriers (Luiu et al., 2017). Policymakers should recognize and address this suppressed travel desire among Hispanic individuals with disabilities. Efforts should center on comprehending the unique challenges confronting this subgroup and tackling the root causes of their restricted travel desires. Engaging with the Hispanic community, including those with disabilities, is imperative to grasp their transportation needs and preferences. This approach will enable policymakers to develop more inclusive transportation policies and programs that cater to the diverse requirements of this population. In the long term, there is a need for more targeted research and data collection focused on racial and ethnic minorities with disabilities. In an immediate way, cultural competency training

should be provided for transportation providers and local agencies to ensure they understand the unique needs and preferences of Hispanic individuals with disabilities in order to improve the quality of transportation services and increase trust among the Hispanic community. This can involve targeted outreach programs, improved accessibility in Hispanic communities, culturally sensitive transportation services, and educational initiatives to raise awareness about travel opportunities and rights. Effective models for engaging minority communities, including Hispanic PWD, in transportation planning and service delivery can be adapted from broader public health or social service outreach. For instance, some Hispanic communities in Texas utilized community health workers as liaisons to bridge trust and communication gaps, a model that could be adapted for transportation navigators. Partnerships with established community-based organizations trusted by the Hispanic community in California have also proven effective in co-designing services (Rodríguez et al., 2024). Developing and sustaining culturally competent transportation services for Hispanic PWD involves overcoming challenges such as securing funding for bilingual staff and translated materials, moving beyond tokenistic engagement to genuine co-creation, addressing potential internal biases within transport agencies, and ensuring that “culturally sensitive” approaches are nuanced and avoid stereotypes, recognizing the diversity within the Hispanic population itself (van Dort, 2019; Ward and Walsh, 2023).

To address the systemic inequities faced by individuals with disabilities, it is crucial to invest in inclusive transportation infrastructure. This includes improving accessibility in public transportation systems (Noor and Iamtrakul, 2023; Blumenberg and Agrawal, 2014), ensuring the availability of specialized transportation options for individuals with disabilities, and creating barrier-free environments that support independent travel. In the long term, efforts should continue to raise awareness and educate the general public about the challenges and needs of individuals with disabilities regarding travel. This can help reduce stigmatization, foster inclusive attitudes, and promote a supportive environment for individuals with disabilities to participate in travel and community activities.

9 Conclusion

Despite increasing awareness of the unmet travel needs of PWD, there is a lack of research that explores the intersectionality of disability with other aspects of an individual's identity, such as gender, race, ethnicity, employment status, income, and age. To bridge this research gap, this study investigated the disparities of the unmet travel needs within specific subgroups of the disabled population using data from the 2017 NHTS. A logit model with interaction effects was developed to identify influential factors contributing to the decision to “reduce day-to-day travel” due to their conditions, as well as to capture the disparities among different groups of PWD.

This study shed light on the specific challenges faced by different subgroups of the disabled population, emphasizing the need for tailored interventions and support to address disparities in travel experiences, addressing a critical gap in the literature where the focus has often been on the general population or on specific aspects of disability without considering the full spectrum of challenges faced by this group. These results highlight the nuanced relationships among gender, poverty status, race, and travel experiences among individuals with disabilities. The findings emphasize the importance of considering specific subgroups within these categories to gain a deeper understanding of the travel challenges and needs faced by individuals with disabilities. Notably, the research highlights the potential suppression of travel desires, and the overlooked travel needs among Hispanic PWD and stay-at-home mothers or wives with disabilities, signaling an urgent call for targeted strategies to improve their access to travel opportunities.

An important aspect of our analysis is the interpretation of reduced travel among PWD. It is crucial to recognize that an indication of reduced travel does not necessarily mean that these individuals made fewer trips or encountered fewer barriers. Instead, it may reflect a suppression of their travel needs due to systemic inequities that limit their opportunities and ability to travel.

For some PWDs, reduced travel might not simply indicate fewer trips but rather a constrained ability to fulfill their desired travel needs. Systemic inequities—such as inadequate transportation infrastructure, lack of accessible services, or economic barriers—can suppress the travel needs of individuals. This suppression leads them to report reduced travel not because their mobility needs are met, but because their travel aspirations are curtailed by external factors. For example, individuals may avoid travel because they have learned that the challenges or barriers they would face make the effort too burdensome. In this context, the absence of reduced travel does not imply satisfaction with their mobility but rather a resignation to the limitations imposed by systemic barriers.

These insights suggest that policymakers and transportation planners need to look beyond simple metrics of travel frequency when assessing the needs of PWDs. It is essential to understand the underlying reasons why some individuals report reduced travel while others do not. Acknowledging that reduced travel might reflect suppressed needs rather than a fulfillment of mobility desires is key to designing more equitable and inclusive transportation policies. For instance, interventions should aim not only to increase the number of trips but also to address the root causes that lead to suppressed travel needs. This could involve improving accessibility, providing more comprehensive support services, and addressing economic barriers that disproportionately affect PWD.

Future studies should explore the nuanced reasons behind reported travel behavior among PWD. Qualitative research could provide deeper insights into how systemic inequities shape travel desires and the perception of travel barriers. Under-

standing these dynamics can help in developing more effective strategies to support the mobility and autonomy of PWD, ensuring that their travel needs are not just met, but fully realized.

The findings also provide valuable insights for service providers, and advocacy groups working to address transportation equity by tailoring interventions to the diverse and intersecting needs of individuals with disabilities, with particular attention to gender dynamics.

This study is subject to a few limitations. Built environment attributes may substantially influence the unmet travel needs of PWD. However, the variables explored in this study did not exhibit significant impacts in the model. This could be due to the aggregated nature of the built environment measures available in the NHTS dataset. Additional measures provided by the smart location database (Sadeh and Tawfik, 2020) could be explored to further examine the impacts of built environment factors. Additionally, this study only focused on whether the respondents reduced travel due to their travel-limiting disability conditions, while there are other compensating strategies that could be taken by PWD, such as asking others for rides, limiting travel to daytime, and using transit more often. Future studies should look into the choices of various options and the influencing factors as well as the potential disparities in their choices.

CRediT authorship contribution statement

Siyu Zhang: Writing – review & editing, Methodology, Formal analysis, Conceptualization, Writing – original draft, Investigation, Data curation. **Md Al Adib Sarker:** Writing – review & editing, Methodology, Writing – original draft, Investigation. **Ibukun Titiloye:** Writing – review & editing, Methodology, Writing – original draft, Investigation. **Xia Jin:** Writing – original draft, Resources, Methodology, Formal analysis, Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The Large Language Model ChatGPT 3.5 was used to improve the grammar, phrasing, and clarity of the manuscript.

Author contributions

The authors confirm contribution to the paper as follows: study conception and design: Siyu Zhang and Xia Jin; data collection: Siyu Zhang; analysis and interpretation of results: Siyu Zhang, Md Al Adib Sarker, Ibukun Titiloye, and Xia Jin; draft manuscript preparation: Siyu Zhang, Md Al Adib Sarker, Ibukun Titiloye, and Xia Jin. All authors reviewed the results and approved the final version of the manuscript.

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